

Notice the function form includes parameters a_1, a_2, a_3 so that we do not force any preconceived notions onto the model such as a square root function with size or a linear relationship with volatility. These parameter values are derived from our underlying data set.

At this point, we alert readers about setting fixed parameter values. For example, many industry participants set $a_2 = \frac{1}{2}$ which is the square root function—or more precisely, numerous industry participants assume that costs increase with the square root of size. Fixing this parameter value to $a_2 = \frac{1}{2}$ implies that investors will react to order and trade information in the same exact manner for all stocks and markets and in all time periods regardless of the underlying economic conditions. Which is simply not true. Investors react to trade information in different ways at different times. For example, recall the differences in market dynamics during the low volatility period of 2003–2004 and the high volatility period of Sept 2008 through Mar 2009.

Where did this $\frac{1}{2}$ or square root belief come from? We believe the $\frac{1}{2}$ power was set in place due to the way volatility scales with time or because of the $\frac{1}{2}$ parameter behind the optimal economic order quantity model. While in these cases there is a natural reason for the $\frac{1}{2}$ parameter, the same is not true when dealing with market impact cost modeling or price evolution and displacement based on order information.

There have been other functional forms of I^* that have been proposed. For example, *Optimal Trading Strategies* (2003) presents three forms of the I-Star model:

$$I_{bp}^* = a_1 \cdot \left(\frac{Q}{ADV} \right) + a_2 \cdot \sigma + a_3 \quad (4.14)$$

$$I_{bp}^* = a_1 \cdot \left(\frac{Q}{ADV} \right) + a_2 \cdot \sigma^{a_3} + a_4 \quad (4.15)$$

$$I_{bp}^* = a_1 \cdot \left(\frac{Q}{ADV} \right)^{a_2} \cdot \sigma^{a_3} \quad (4.16)$$

We have performed significant testing on these models (as well as other function forms) using numerous data sets, time periods, and global regions, and found the power function formulation to be the most robust, stable, and accurate.

One important question that surfaces: Since the parameters a_1, a_2, a_3, b_1 are estimated across a data set of stocks and are identical for all stocks, how do we differentiate trading cost across different stocks and the same order size?